

The DELPH-IN Grammarly A Year of New Interfaces

What's new since DELPH-IN Summit 2025

Francis **Bond**

Computational Linguistics Lab

Asian Studies & Sinofon Center & General Linguistics

Palacký University, Olomouc, Czechia

<bond@ieee.org>

DELPH-IN Summit 2026



Faculty
of Arts

Last Year, and This Year

- DELPH-IN 2025: introduced *the Codex* — one archive for every DELPH-IN grammar, compiled under ACE, loaded into the LTDB
- Ended with a To Do list, including:
 - ▶ “Look at full tree search over trees and DMRS – Grew-match”
 - ▶ “I desperately need someone who understands JavaScript...”
- This talk: what got done about that list



Roadmap

- 1 The Grammarly, Briefly
- 2 Try It: The Interactive Demo
- 3 Rendering in the Browser
- 4 A Static Mirror for the LTDB
- 5 Structural Search with Grew-match
- 6 The ERS Catalogue, Made Executable
- 7 Wrap-up



Recap: One Home for 23 Grammars

- 23 HPSG grammars, 17 languages, 8 families
- Standardized metadata, compiled under ACE, loaded into the LTDB
- FAIR4RS-compliant, archived on Zenodo, annual releases timed to the Summit
- Now called the **Grammar**
(*grammar*, *gramarye*, n. — archaic: mystical learning; grammar)
- <https://github.com/delph-in/grammar>
DOI: 10.5281/zenodo.18945956
- Live: <https://compling.upol.cz/ltdb>



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Parse and Generate, No Install Required

- Built on pydelphin (Goodman, 2019)
- Type a sentence, see the derivation tree and semantics
- Every type, rule, and predicate links back to its LTDB entry
- Where the grammar supports generation (ERG and others): pick a parse, inspect its MRS, generate surface strings from it

Grammars ERG_2025.db Rules Lexical Types Demo lemma/type/predicate 🔍

☒ Tree ☒ MRS ☒ DMRS

1 result(s) for: *I stand on the shoulders of giants.*

Result 1

```
graph TD
    sbhd[sb-hd mc_c] --> hdnbnpc[hdn_bnp-qnt_c]
    sbhd --> hdaj[hd-aj_int-unsl_c]
    hdnbnpc --> i[i]
    hdaj --> vn3s[v_n3s-bse_illr]
    hdaj --> hdcmp1[hd-cmp_u_c]
    vn3s --> stand1[stand1]
    stand1 --> stand[stand]
    hdcmp1 --> on[on]
    hdcmp1 --> sphd[sp-hd_n_c]
    on --> on_text[on]
    sphd --> the1[the_1]
    the1 --> the[the]
    sphd --> hdnaj[hdn-aj_redrel_c]
    hdnaj --> npl[n_pl_olr]
    npl --> shldr[shoulder_n1]
    shldr --> shldr_text[shoulders]
    hdnaj --> hdcmp2[hd-cmp_u_c]
    hdcmp2 --> ofpos[of_poss]
    ofpos --> of[of]
    hdcmp2 --> hdnbnpc2[hdn_bnp_c]
    hdnbnpc2 --> hdpct[hd-pct_c]
```



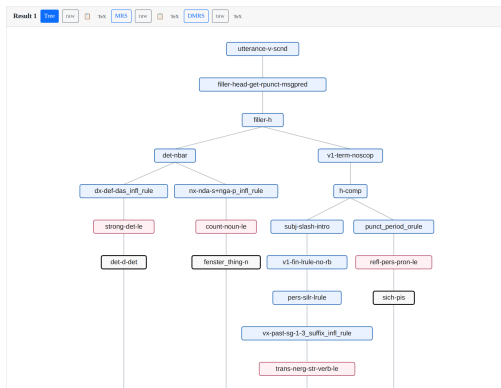
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Derivation Trees, Laid Out in JavaScript

- Contour-based layout (Reingold–Tilford tradition), bottom-up with a running cursor
- Toggle: grammar rule names $\leftrightarrow$ phrase-structure labels (ace -L)
- Parsed with --udx=all: lexical type and lexical entry shown above every word, not just the surface form
- Collapsible nodes; shift-click a node to follow its type link



MRS and DMRS, Also in the Browser

- MRS: interactive attribute-value matrix — click a variable, every co-indexed occurrence highlights
- DMRS: MRS-to-DMRS conversion reimplemented in JavaScript, full parity with pydelphin; rendered as an SVG graph
- Non-standard MRS extensions (GG, Norsyng) fall back to the raw ACE string gracefully

```
TOP: h0 INDEX: e2 [e]

_def_q_rel<0:3> LBL:h4 ARG0:x9 [x]
--DIND:i5 WLINK:*List* CFROM:"0" CTO:"3" RSTR:h10 body:h11

_fenster_n_rel<4:11> LBL:h12 ARG0:x9 [x]
--DIND:i13 WLINK:*List* CFROM:"4" CTO:"11" --rel:i17 --relh:h18

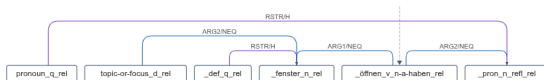
_offnen_v_n-a-haben_rel<12:19> LBL:h1 ARG0:e2 [e]
--DIND:i19 WLINK:*List* CFROM:"12" CTO:"19" ARG1:x9 ARG2:x23

_pron_n_refl_rel<20:26> LBL:h24 ARG0:x23 [x]
--DIND:i25 WLINK:*List* CFROM:"20" CTO:"26" --rel:i29 --relh:h30

pronoun_q_rel<-1:-1> LBL:h31 ARG0:x23 [x]
--DIND:i32 WLINK:*List* CFROM:*top* CTO:*top* RSTR:h36 body:h37

topic-or-focus_d_rel<-1:-1> LBL:h1 ARG0:e42 [e]
--DIND:i38 WLINK:*List* CFROM:*top* CTO:*top* ARG1:e2 ARG2:x9

HCONS: h0 qeq h1, h10 qeq h12, h36 qeq h24
```



One Click to L^AT_EX

- Tree → forest
- MRS → langsci-avm
- DMRS → tikz-dependency
- A single copy from the demo into a paper draft



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Why a Static Mirror?

- The live LTDB is the richest interface — and a single point of failure: Flask app, grammar databases, parser binaries, sysadmin
- Dynamic, database-driven content decays faster than static files: link-rot studies put it at roughly a fifth of all failures (Sadatmoosavi et al., 2026)
- Goal: keep grammars inspectable even when the server is down
- First attempt (Flask-Freeze, one file per type) does not scale: 129,526 type pages, ~ 1.8 GB for the ERG alone



Compact Per-Grammar SQLite + sql.js

- Hybrid design: one static shell, tiny per-grammar databases loaded and queried client-side
- Type hierarchy, definitions, docs, source locations, frequencies, and a curated set of examples

Grammar	Source DB	Compact type DB	Compact gzip
ERG	~2.0 GB	5.2 MB	1.3 MB
HeGram	183 MB	9.9 MB	3.1 MB

- Live parsing, generation, and full-text search stay server-side



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Off Last Year's To Do List

- Search was type names, lemmas, predicates only — no way to query *structure*
- Rather than build a bespoke engine: reuse Grew, the graph matcher built for UD (Guillaume, 2021)
- New tool, db2grew.py: derivation trees → constituency-style graphs; DMRS → dependency-style graphs
pattern { V -[1=ARG1]-> X; V -[1=ARG2]-> X }

The screenshot shows the DELPH-IN Grammar Explorer interface. At the top, there's a search bar with 'ERG: 2025: dltres' and a 'DELPH-IN Grammar Explorer' header. Below the search bar, there's a code editor showing a query for 'retire' with its arguments. To the right, there's a sidebar with 'DMRS', 'Trees', 'ERS', and 'Metadata' tabs. The main area displays '492 occurrences (0.784s)' for the query. Below this, there's a list of occurrences and a DMRS graph for the sentence 'Every retiring person has a different situation facing him.' The graph shows the derivation of the sentence from the root node 'every_q' through various grammatical nodes like 'retire_v_1', 'person_n_1', 'have_v_prd', 'a_q', 'different_a_than_from', 'situation_n_1', 'face_v_1', 'pronoun_q', and 'pronoun_n'. The graph is a dependency-style graph showing the relationships between the words in the sentence and their grammatical roles.

One Command, No Separate Setup

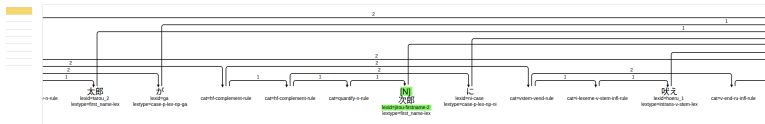
- Export, corpus merge, and backend precompilation folded into the Grammary build
- Starting Grew-match alongside the LTDB: one command, no manual server juggling
- Full ERG 2025 treebank — 97,650 sentences, ~195,000 graphs: export in under 30 seconds, Grew-match compiles and serves in under two minutes
- Snippet library ships with the tool: worked queries by corpus type, so newcomers start from a working pattern



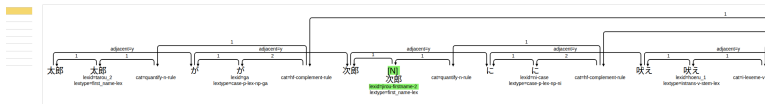
Making a Dependency Tool Draw Trees

- Grew-match's renderer, dep2pict, has no constituency mode — confirmed by tracing the source, not just the docs
- Trick: give *every* node a position, in postorder, not just the surface tokens
- Result: arcs nest by constituent structure, no crossings
- Cost: adjacent words are no longer adjacent positions — restored with an explicit adjacent=y edge per word pair

太郎 が 次郎 に 映え させる つもり だった 。



太郎 が 次郎 に 映え させる つもり だった 。



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26 Phenomena, Released as Queries

- ERG Semantics Documentation: each phenomenon has an ERS fingerprint (Flickinger et al., 2014)
- All 26 — apposition to vocatives — translated into Grew queries over the exported DMRS corpora
- Every query returns matches on the ERG 2025 treebank; a checker recounts them against any corpus Grew-match serves
- Doubles as a cheap semantic regression signal after a grammar or treebank update
- An ERS tab in the snippet pane lists every phenomenon, straight from the same TOML file the checker reads


DOLPHIN Grammar Colors
ERC 2025 dmrs

ERC 2025 dmrs

```

1  <A fall 0> is followed by another nominal phrase with the same referent: the
2  <operation antecedent> in two places: <fall on the two>
3
4  @em: <Argues> [1] [1] [RefOn on githh, <ndv/erg/Transitive, Agentive>]
5  @em: <Argues> [1] [1] [RefOn on githh, <ndv/erg/Transitive, Agentive>]
6  @em: <Argues> [1] [1] [RefOn on githh, <ndv/erg/Transitive, Agentive>]
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8  pattern: <A [tresh+zero]> <A [tresh+zero]> <X> <A [tresh+zero]> <Y>
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What We Did This Year

- Interactive parse-and-generate demo — no install required
- Client-side rendering of trees, MRS, DMRS, with one-click $\text{\LaTeX}$ export
- Static SQLite/sql.js mirror: the ERG down from ~ 2 GB to 5.2 MB
- Grew-match integration: db2grew.py, single-command build, tree-legible rendering
- 26 ERS phenomena released as executable, checkable queries



Try It

- Live demo: <https://compling.upol.cz/ltdb>
- Grammarly: <https://github.com/delph-in/grammarly>
- Zenodo: [10.5281/zenodo.18945956](https://zenodo.org/record/18945956)
- Questions, requests, missing grammars — let us know



References I

- Flickinger, D., Bender, E. M., and Oepen, S. (2014). Towards an encyclopedia of compositional semantics: Documenting the interface of the English Resource Grammar. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, pages 875–881, Reykjavik, Iceland. European Language Resources Association (ELRA).
- Goodman, M. W. (2019). A python library for deep linguistic resources. In *2019 Pacific Neighborhood Consortium Annual Conference and Joint Meetings (PNC)*, Singapore.
- Guillaume, B. (2021). Graph matching and graph rewriting: GREW tools for corpus exploration, maintenance and conversion. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 168–175, Online. Association for Computational Linguistics.

Sadatmoosavi, A., Khasseh, A. A., and Tajedini, O. (2026). Link rot in LIS literature: A 20-year study of web citation decay, recovery and preservation challenges. *Aslib Journal of Information Management*.